

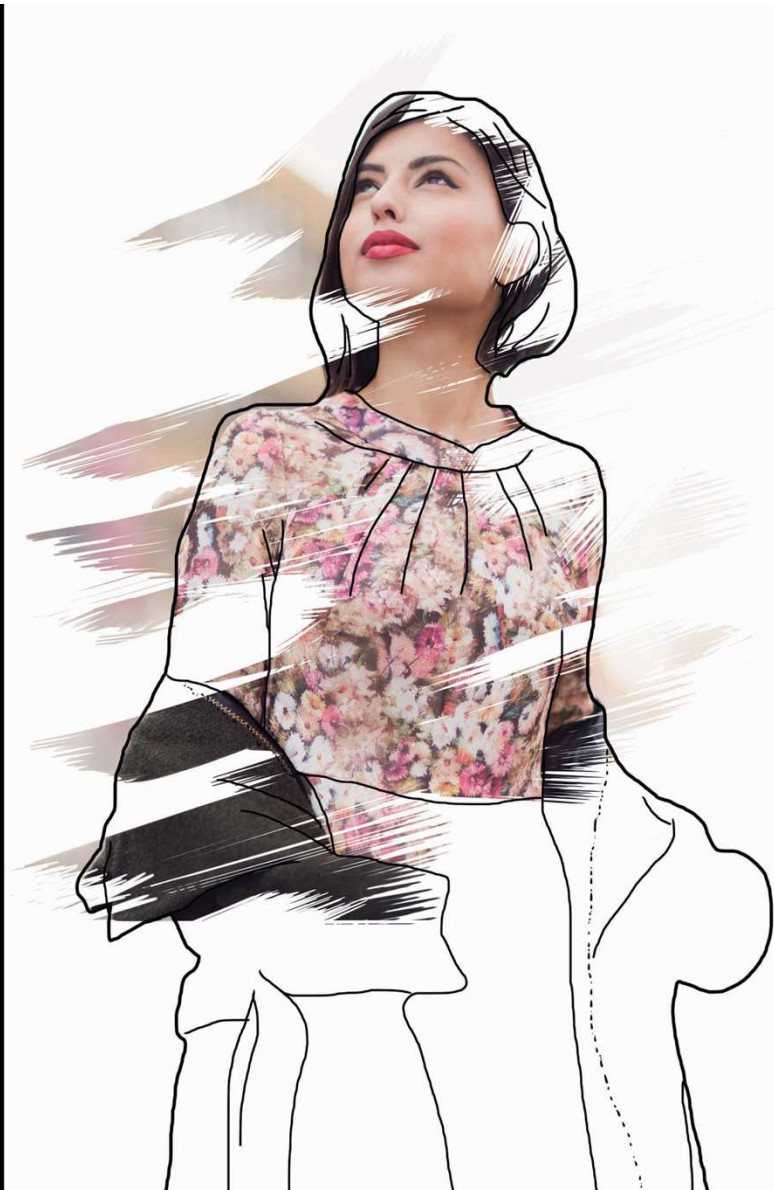
HALLSTAR 

B E A U T Y

BRIGHT Oléoactif®

Global complexion control

HALLSTAR ✦
B E A U T Y

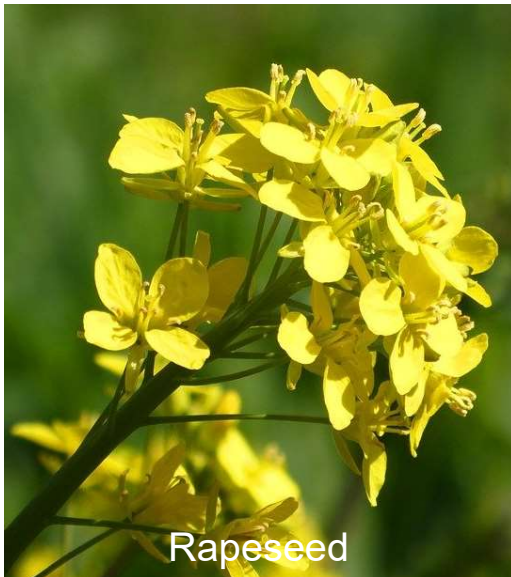


Extraction Process and Composition



**Oléo-éco-
extraction**

- Green technology
- Natural and bioactive extraction solvent: oil and glycerides from **Rapeseed**
- Safe (no organic solvent traces)
- Plants with brightening properties: **Marshmallow** (*Althaea officinalis* L.), **Rice bran** (*Oryza sativa* L.), **Licorice** (*Glycyrrhiza glabra* L.)



Natural Rapeseed Solvent

Rapeseed oil



- European origin
- Organic quality
- Harvest in July
- Independent French company, human scale, ethical and responsible practices
- Cakes (by-products of oil extraction) used as animal feed



- **Repairs, nourishes** skin (w9)
- Promotes cell communication, moisturizes, reinforces barrier function of epidermis and **boosts skin radiance** (w6)
- **Anti-inflammatory and soothing** (w3)
- Antioxidant tocopherols

Main fatty acids

Oleic acid (w9): 55-65 %

Linoleic acid (w6): 20-25 %

α -Linolenic acid (w3): 8-11 %



Organic Rhizome and Root

Marshmallow rhizome



- European origin
- **Organic** quality
- Harvest in fall
- Brightening agent: inhibition of endothelin-1 which inhibits melanocyte proliferation and differentiation, **inhibits the pigmentation process**

Licorice root



- European or African origin
- **Organic** quality
- Harvest in fall
- Independent French company with sustainable development practices
- Brightening agent: inhibition of tyrosinase which **inhibits melanin synthesis**



Upcycled Rice Bran

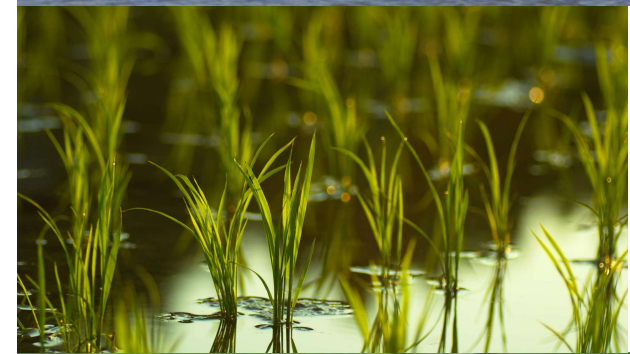
Camargue rice bran



- **French** origin
- **Upcycled**
- PGI (Protected Geographical Indication) label
- Waste (rice straw and husks) recovery in house insulation and animal litter



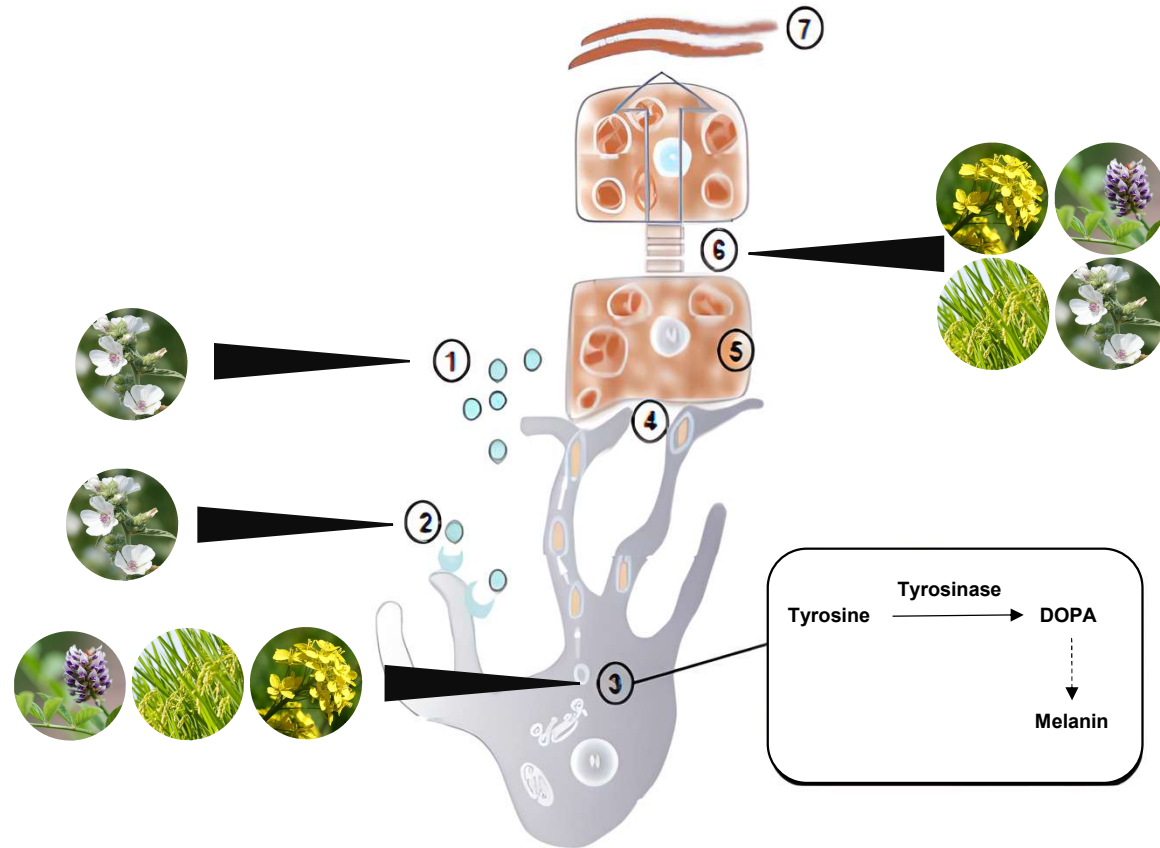
- **Inhibition of tyrosinase** activity and reduction of **melanogenesis gene expression**



Mechanism of Action BRIGHT Oléoactif®

- ① Endotelin-1 secretion and release by the keratinocytes
- ② Triggering of melanogenesis by the activation of endothelin-1 on the melanocytes / Melanocytes differentiation and proliferation
- ③ **Synthesis** of melanin : activation of tyrosinase enzyme
- ④ **Transport** of melanin into keratinocytes
- ⑤ **Coloration** of melanin (oxidation)
- ⑥ **Migration** of melanin
- ⑦ **Accumulation** of melanin in stratum corneum pigmentation of the skin

Thanks to its composition, BRIGHT Oléoactif® regulates skin pigmentation disorders through **complementary pathways**.



Comparison with Whitening References

① KOJIC ACID



② HYDROQUINONE Oc1ccc(O)cc1

BRIGHT Oléoactif® is of vegetable origin.

A dose of 0,12 % of BRIGHT Oléoactif® inhibits ~ 90 % of tyrosinase activity which is equivalent to 0,01 % of pure hydroquinone.

It can be used in all cosmetics including daily care and sun care products for all skin types, even the more delicate.

Compound	Negative aspects
Kojic acid	Sensitizing, carcinogenic potential
Hydroquinone	Cytotoxic, carcinogenic potential
Arbutin	Hydroquinone precursor, expensive
Licorice extract (glabridin)	Limited efficacy, expensive, difficult to formulate
Ascorbic acid	Low efficacy, unstable

The combination of the **3 plants** and the **oil-based form** makes it **safe and an original alternative** to kojic acid and hydroquinone.



Comparison with Whitening References

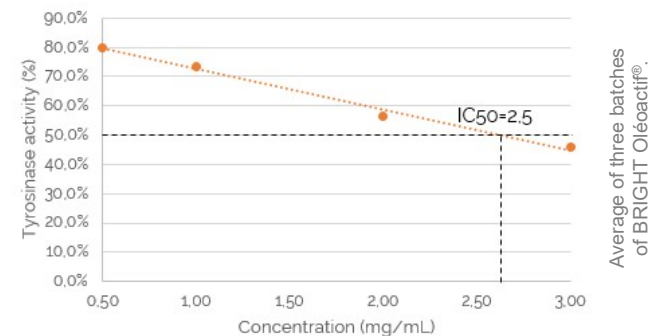
3 ARBUTIN

In the range of tested concentrations, BRIGHT Oléoactif® inhibits significantly, and in a dose dependent manner, tyrosinase activity. Its IC₅₀ (2,5 mg/mL) is comparable to that of arbutin (2,7 mg/mL). But this reference molecule is not always well-tolerated by sensitive skin.

BRIGHT Oléoactif® is **as effective as** arbutin **with a higher tolerance.**

Compound	Negative aspects
Kojic acid	Sensitizing, carcinogenic potential
Hydroquinone	Cytotoxic, carcinogenic potential
Arbutin	Hydroquinone precursor, expensive
Licorice extract (glabridin)	Limited efficacy, expensive, difficult to formulate
Ascorbic acid	Low efficacy, unstable

Inhibition of tyrosinase of Bright Oléoactif®





Comparison with Whitening References

4 GLABRIDIN

Licorice whitening extracts are used at very low concentrations but are very expensive and difficult to formulate. Used at the same concentration, BRIGHT Oléoactif® provides better efficacy, is cost effective and easy to formulate.

BRIGHT Oléoactif® is **more effective** than pure glabridin (+ 24 %).

Compound	Negative aspects
Kojic acid	Sensitizing, carcinogenic potential
Hydroquinone	Cytotoxic, carcinogenic potential
Arbutin	Hydroquinone precursor, expensive
Licorice extract (glabridin)	Limited efficacy, expensive, difficult to formulate
Ascorbic acid	Low efficacy, unstable



Comparison with Whitening References

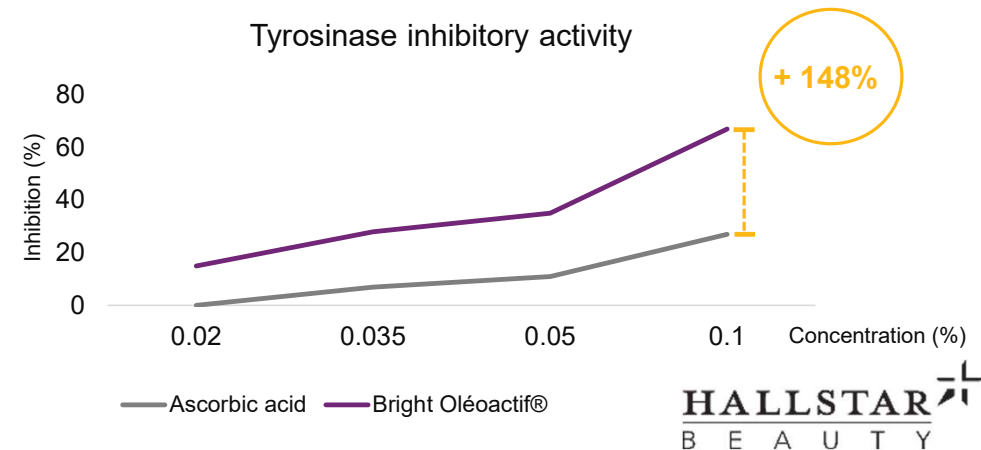
5 ASCORBIC ACID

The tyrosinase inhibitory activity of BRIGHT Oléoactif® is significantly higher than vitamin C efficacy.

Furthermore it is really stable.

At 0.1% dose, BRIGHT Oléoactif® is **more than twice as effective as** ascorbic acid.

Compound	Negative aspects
Kojic acid	Sensitizing, carcinogenic potential
Hydroquinone	Cytotoxic, carcinogenic potential
Arbutin	Hydroquinone precursor, expensive
Licorice extract (glabridin)	Limited efficacy, expensive, difficult to formulate
Ascorbic acid	Low efficacy, unstable



Comparison with Whitening References

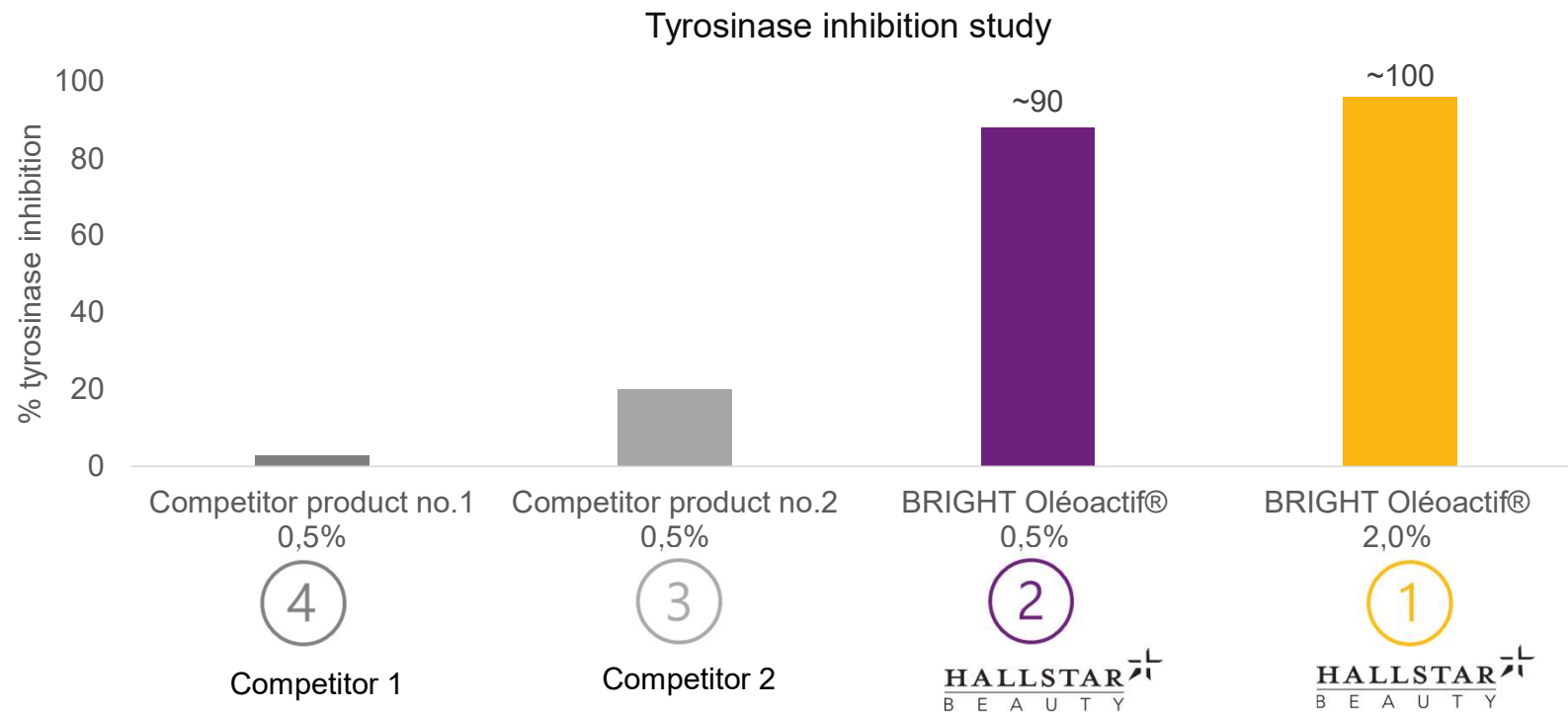
BRIGHT Oléoactif® is **more effective than current skin lighteners**. Moreover, it has **no side effects**, unlike some of these other actives.

Compound	Negative aspects
Kojic acid	Sensitizing, restricted use
Hydroquinone	Cytotoxic, carcinogenic potential
Arbutin	Hydroquinone precursor, expensive
Licorice extract (glabridin)	Limited efficacy, expensive, difficult to formulate
Ascorbic acid	Low efficacy, unstable
BRIGHT Oléoactif®	None !





Comparison with Competitors



0,5% BRIGHT Oléoactif® is **more effective than other skin whitening actives** tested with the same concentration. Furthermore, results show a **dose-dependent inhibition** of tyrosinase activity.

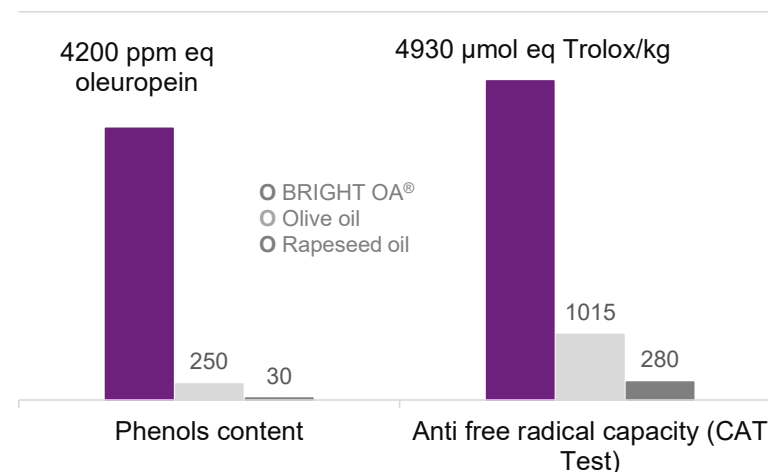
Inhibition of Melanin Coloration



- Free radicals exposures (for example UV) are responsible of immediate skin coloration due to the oxidation of existing pigments and to a redistribution of melanosoms.
- Pre-existing pigmentation skin disorders may be aggravated by that kind of exposure.

Thanks to its **anti-free radical activity**, BRIGHT Oléoactif[®] avoid melanin coloration.

The clinical study will confirm these data.



- Phenols content : average content of 6 representative batches.
- CAT Test Method : generation of free radicals by AAPH and measure by spectrophotometry of the ability to block oxidative radical reactions.

Clinical Efficacy

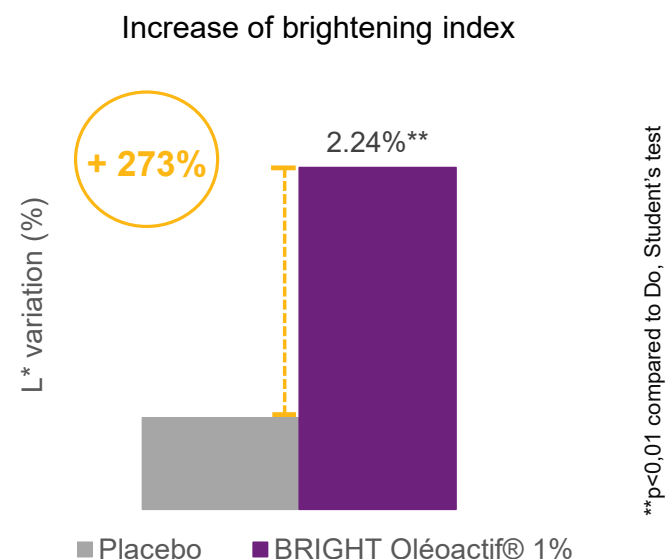


LABORATOIRE
BIOEC

1 BRIGHTENING EFFICACY

- **20 Korean women** with age spots applied on their faces 1% BRIGHT Oléoactif® cream versus placebo without active, twice a day, over a period of 56 days.
- **Evaluation of skin brightness** by objective spectrophotometric measurement.
- **Analysis of the L* value.** This value corresponds to the luminosity of the skin (clarity, luminance). It can be related to the global brightening efficacy of a product. The more L* is increased, the brighter the skin.
- This value has been measured at day 28 and compared to the placebo data.
- Standardized face pictures : full face analysis with VISIA-CR.

1% BRIGHT Oléoactif® **increases skin brightening index L* by 273%** compared to placebo after only 28 days. The skin is visibly brighter.



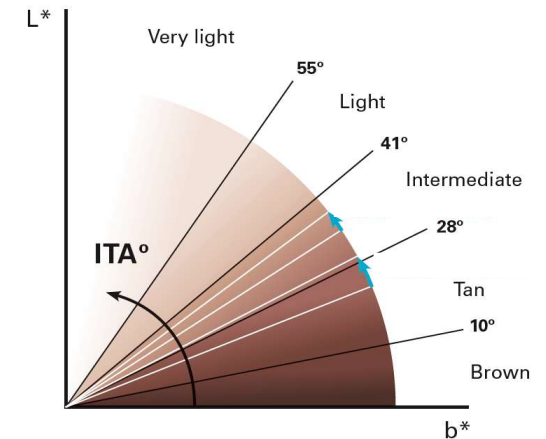
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Clinical Efficacy



2 WHITENING EFFICACY

- 20 Korean women with age spots applied on their faces 1% BRIGHT Oléoactif® cream versus placebo without active, twice a day, over a period of 56 days.
- Evaluation of skin whiteness by objective spectrophotometric measurement.
- Analysis of the **Individual Typological Angle** (ITA). This value corresponds to the skin pigmentation degree. It can be related to the whitening efficacy of a product. The more ITA is increased, the whiter is the skin.
- This value has been measured at day 28 and day 56. For each woman, measurements have been performed **on a non-spotted zone** located near a spotted zone.
- Results were compared to the placebo data.
- Standardized face pictures : full face analysis with VISIA-CR.

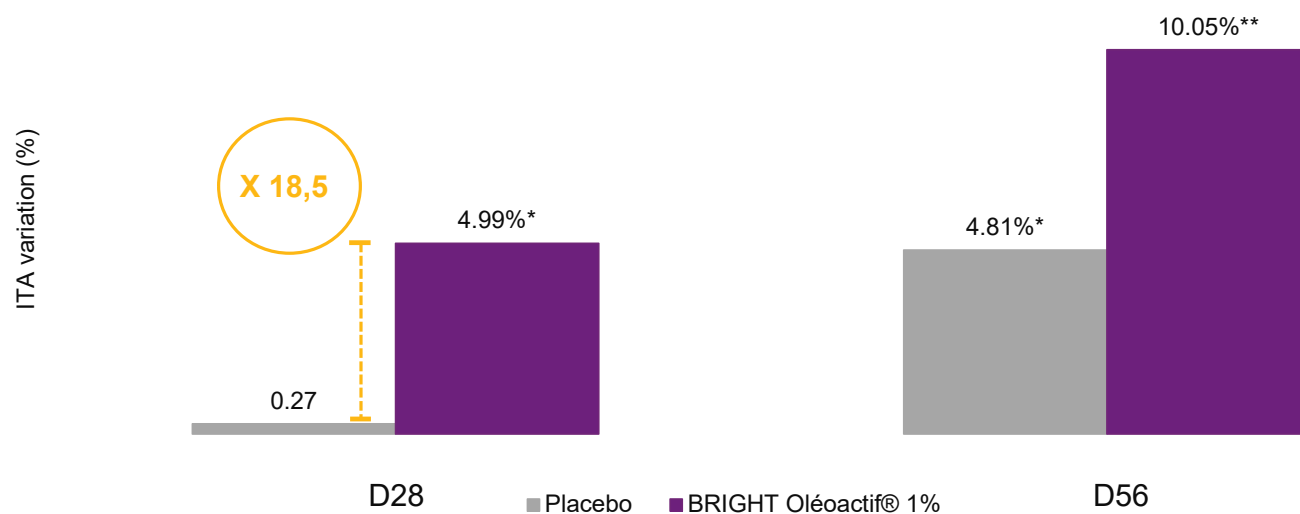


Clinical Efficacy



2 WHITENING EFFICACY

Increase of whitening index on non-spotted zone



*p<0,05 compared to Do, Student's test
**p<0,01 compared to Do, Student's test

After **28 days** of treatment, the significant increase of ITA correlates with a **significant decrease in skin color**. After 56 days, the results are even better.

Clinical Efficacy



3 VISUALISATION OF GLOBAL COMPLEXION CONTROL

Full face analysis with VISIA-CR - Skin pigmentation of a volunteer before and after 1% BRIGHT Oléoactif® treatment.



D0



D28



D56

BRIGHT Oléoactif® offers **overall brightening** and **homogenization of the skin**.

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Clinical Efficacy

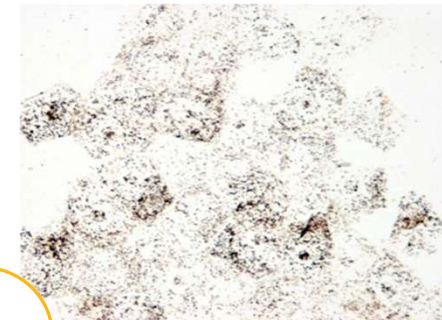


4 ANTI DARK SPOT ACTION:

- 20 Korean women with age spots applied on their faces 1% BRIGHT Oléoactif® cream versus placebo without active, twice a day, over a period of 56 days.
- Corneocytes collection by D-Squame stripping discs. Coloration of melanin.
- Dosage of the skin melanin content at day 56, compared to the placebo data.
- This investigation method is perfectly suited to the specific evaluation of the efficacy of keratosis (thickening of the skin's stratum corneum and high level of melanin in upper layers).

1% BRIGHT Oléoactif® decreases the synthesis and the coloration of melanin in the cells. The quantity of **melanin in dark spots** is **visibly decreased (-52%)**.

Effect on melanin content in dark spots



Placebo
D56



1% BRIGHT
Oléoactif®
D56

- 52%

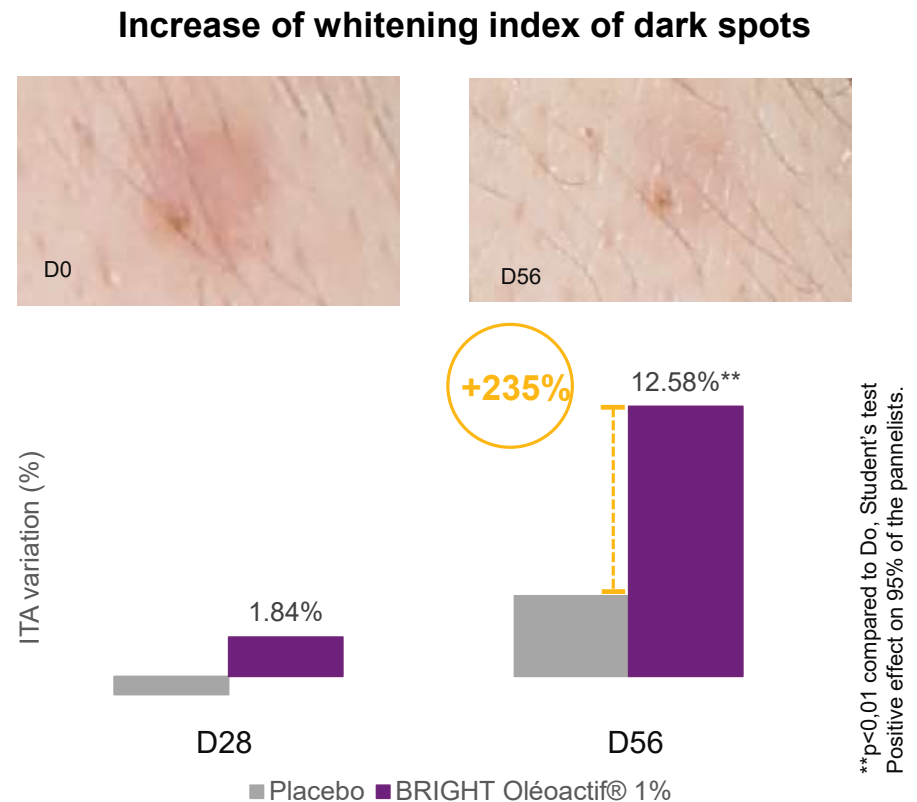
Clinical Efficacy



4 ANTI DARK SPOT ACTION:

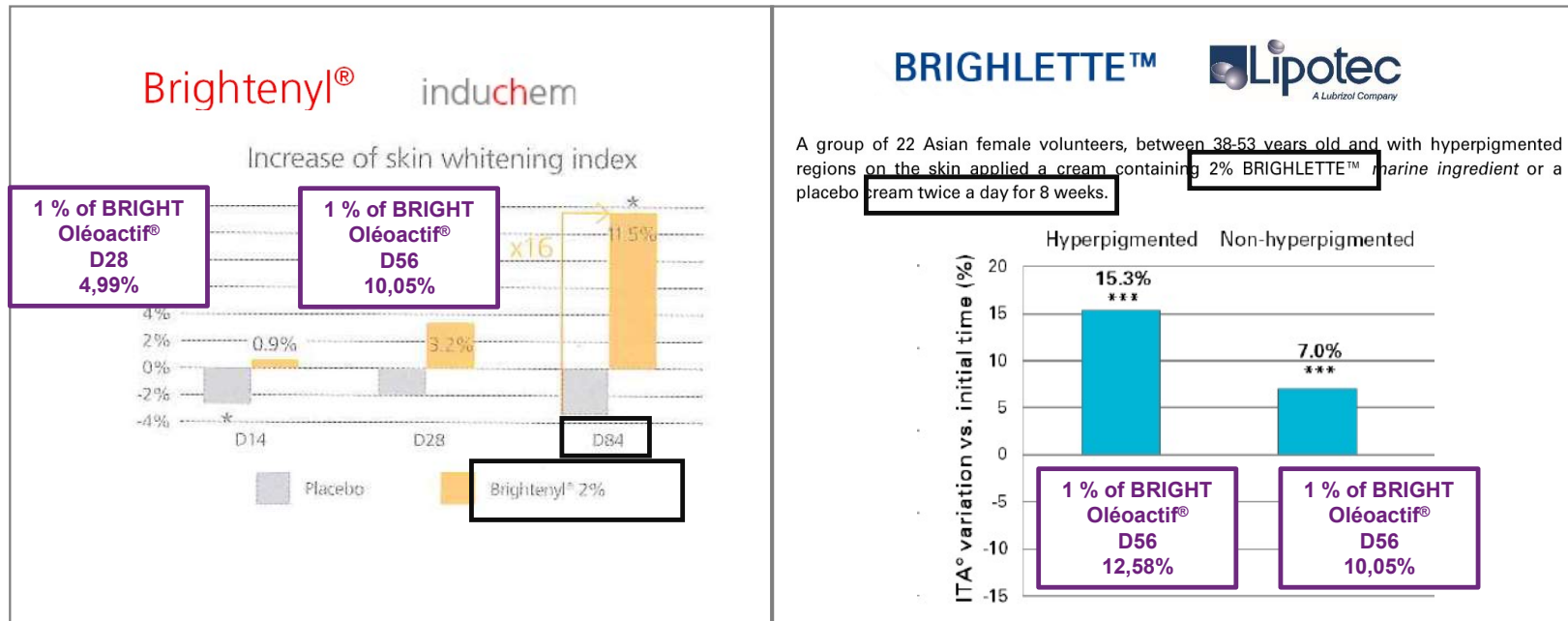
- Analysis of the Individual Typological Angle (ITA). This value corresponds to the skin pigmentation degree. It can be related to the whitening efficacy of a product. The more ITA is increased, the whiter the skin.
- This value has been measured at day 28 and day 56. For each woman, measurements have been performed **on a spotted zone**.
- Standardized pictures of specific dark spot areas.

1% BRIGHT Oléoactif® is effective in **reducing the appearance of dark spots**. Visible signs of color reduction are observed after 56 days.



Clinical Efficacy

COMPETITOR BENCHMARKING



BRIGHT Oléoactif® is a powerful skin lightener !

Technical and Regulatory Data

INCI name	Brassica Campestris (Rapeseed) Seed Oil (and) Glycyrrhiza Glabra (Licorice) Root Extract (and) Polyglyceryl-3 Diisostearate (and) Althaea Officinalis Root Extract (and) Oryza Sativa (Rice) Bran Extract
Appearance	Homogenous, liquid, from yellow to yellow-orange
Solubility	Liposoluble
pH	3 - 10
Allergens	None
Whitening status in Asia	• CHINA: can be used in cosmetic, but NOT in Special use Cosmetics. Not registered by the CFDA as a whitening active. • JAPAN: can be used in cosmetic products but NOT for Quasi-Drug. Not registered. • KOREA: can be used in cosmetic, but NOT in functional cosmetic
Dosage	0.5% - 5.0%
Organic certification	84.0% according to COSMOS standard



Claims and Applications



- **Skin brightening**
- **Hyperpigmentation reduction**



- Brightening skin care
- Whitening skin care
- Reduction of hyperpigmentation
- Global anti-aging
- Sun care products



References

- [1] Whelan, J., & Fritsche, K. (2013). Linoleic acid. *Advances in Nutrition*, 4, 311-312.
- [2] Ando, H., Watabe, H., Valencia, J. C., Yasumoto, K. I., Furumura, M., Funasaka, Y., ... & Hearing, V. J. (2004). Fatty acids regulate pigmentation via proteasomal degradation of tyrosinase: a new aspect of ubiquitin-proteasome function. *Journal of Biological Chemistry*, 279, 15427-15433
- [3] Kobayashi, A., Hachiya, A., Ohuchi, A., Kitahara, T., & Takema, Y. (2002). Inhibitory mechanism of an extract of *Althaea officinalis* L. on endothelin-1-induced melanocyte activation. *Biological and Pharmaceutical Bulletin*, 25, 229-234.
- [4] Nerya, O., Vaya, J., Musa, R., Izrael, S., Ben-Arie, R., & Tamir, S. (2003). Glabrene and isoliquiritigenin as tyrosinase inhibitors from licorice roots. *Journal of agricultural and food chemistry*, 51, 1201-1207.
- [5] Jun, H. J., Lee, J. H., Cho, B. R., Seo, W. D., Kang, H. W., Kim, D. W., ... & Lee, S. J. (2012). Dual inhibition of γ -oryzanol on cellular melanogenesis: Inhibition of tyrosinase activity and reduction of melanogenic gene expression by a protein kinase a-dependent mechanism. *Journal of Natural Products*, 75, 1706-1711.



Thank You

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